

CSE340: Computer Architecture

Assignment 4

Chapter 4

Question 01

Design a 16 X 132 register file.

Note: You must label all the input and output pins

Question 02

Why does the ALU control unit take the following bits as input?

Instruction[30, 14-12]

Question 03

What is the significance of the Zero pin in the ALU?

Question 04

Suppose you have a single instruction;

Add X21, X22, X23

Register read = 20s

Instruction Fetch = 20s

Memory Operation = 40s

Register write = 20s

ALU Operation = 40s

Suppose you have a pipelined setup and a single-cycle setup,
Which one will run the instruction faster?

Question 05

Modify the BEQ datapath so that it works for BNE.